

ML2++ Use Case 2 - Smart Home Energy Dataset

Processed file: use_case_2_smart_home_energy.csv

Observations: 4,200 one-minute values

Period: 2016-01-01 05:00:00 through 2016-01-04 02:59:00 UTC

Target: use_kW (household power use, kW)

Source variable: use [kW]

Transformation: arithmetic mean of each consecutive 60 one-second target observations

Raw target observations used: first 252,000

Range: 0.000800 to 4.302675 kW

Mean: 0.775213 kW

Missing output timestamps: 0

Missing target values: 0

Duplicate output timestamps: 0

Documented forecast horizon: 3 minutes

Documented models: ARIMA(1,1,1) and additive Holt-Winters

Provenance

Original source: Kaggle Smart Home Dataset with weather Information by Taranvee. Reconstruction uses the public Yotahub/Machbase converted stream; the repository README links the conversion script and records a SHA-256 digest of the selected 252,000 target pairs.

Selected-pair SHA-256: fd0e874c6c7766fec60bd7c4f2b697ace85e0d558bba33a9dc2fad29b8ef3801

Correction note

Corrected on 13 August 2026. The previously deposited CSV had been derived from humidity while labelled use_kW. It is superseded by the dataset rebuilt from the actual use [kW] column. Results from the superseded CSV require rerunning before being described as household-energy forecasting results.